

## Mean Decrease Impurity (MDI)

How it works?

- node level: every time a decision tree splits data using a feature, impurity of child node decreases
- tree level: algorithm calculates total decrease in impurity by that feature across whole tree
- forest level: total across ensemble of many trees

## Critical considerations

- Masking effect: if one feature is slightly better than another, model might always choose the better therefore masking usefulness of second feature
  - ↳ fix: use `max_features = 1` to force each node to look at only one randomly selected feature
- IS bias: MDI is entirely calculated on training data
- Algorithm specific: MDI relies on decision tree splits. Cannot use this method on SVMs, Logistic regression, Neural networks
- Normalized scores: MDI scales all feature importance scores so the sum of all importance is 1.
- Substitution effect (Collinearity): if you have two correlated features, MDI won't choose but split importance between them.
  - ↳ can make crucial features look unimportant
- Bias toward high-cardinality features: MDI naturally favors continuous variables or categorical variables with high cardinality because tree is more likely to find split points.